

Homework 2

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1 Problem 1

The reaction equations can be written as

$$\begin{aligned}\frac{dA}{dt} &= -2k_1A^2 + 2k_2B - k_3A + k_4C^2, \\ \frac{dB}{dt} &= k_1A^2 - k_2B, \\ \frac{dC}{dt} &= k_3A - k_4C^2.\end{aligned}$$

I first integrated the system using the adaptive RK45 method from Homework 1. The density curves stopped noticeably changing once the relative tolerance was reduced to approximately 10^{-4} and the absolute tolerance to approximately 10^{-5} . Smaller values gave essentially the same visible density curves. However, in order to use the RK45 solution as the reference for the 10^{-6} relative-error condition in the backward Euler comparison, I reduced the tolerances further to 10^{-6} and 10^{-8} , respectively.

For the semi-implicit backward Euler method, the Jacobian of the system is

$$J = \begin{pmatrix} -4k_1A - k_3 & 2k_2 & 2k_4C \\ 2k_1A & -k_2 & 0 \\ k_3 & 0 & -2k_4C \end{pmatrix}.$$

The semi-implicit update was calculated from

$$(I - hJ)\Delta y = hf(y),$$

with

$$y_{n+1} = y_n + \Delta y.$$

Using 1000 backward Euler steps over the full 10 Myr interval, the individual backward Euler points agreed well with the RK45 solution where the backward Euler solution was sampled. The fast molecular hydrogen evolution is not resolved by the large fixed timestep, but the method remains stable and reaches the same equilibrium behavior.

If only the equilibrium densities at 10 Myr are required, the backward Euler method can use much larger timesteps. I found that the minimum number of steps required to keep the relative error below 10^{-6} for all three species was

$$\boxed{11 \text{ steps}}.$$

The relative errors at 10 Myr were

$$\begin{aligned}\epsilon_A &= 1.51 \times 10^{-9}, \\ \epsilon_B &= 3.37 \times 10^{-7}, \\ \epsilon_C &= 7.50 \times 10^{-7}.\end{aligned}$$

The ionized hydrogen density had the largest error and therefore set the limiting timestep.

See `problem1.ipynb` or `problem1.pdf`.

2 Problem 2

The Sturm–Liouville problem was

$$y''(x) + (\lambda\pi)^2 \rho(x)y(x) = 0$$

with boundary conditions

$$y(0) = 0, \quad y(1) = 0.$$

To integrate the second-order equation with RK4, I rewrote it as three first-order equations using the state vector

$$\mathbf{y} = \begin{pmatrix} y \\ y' \\ \lambda \end{pmatrix}.$$

The resulting system is

$$\begin{aligned}y' &= v, \\v' &= -(\lambda\pi)^2 \rho(x)y, \\ \lambda' &= 0.\end{aligned}$$

The two density profiles used were

$$\rho(x) = 1$$

and

$$\rho(x) = 1 + e^{10x}.$$

The boundary value problem was solved with the shooting method. For each trial value of λ , the system was integrated from $x = 0$ to $x = 1$ using RK4. The value of $y(1)$ was then used as the score,

$$F(\lambda) = y(1, \lambda),$$

so that an eigenvalue corresponds to a root satisfying

$$F(\lambda) = 0.$$

The root was first bracketed by increasing the trial eigenvalue and then found using bisection.

For the flat density case, $\rho(x) = 1$, the analytical eigenfunctions are

$$y_n(x) = \sin(n\pi x),$$

with eigenvalues

$$\lambda_n = n.$$

Using initial guesses of 0.5 and 1.5, the numerical solver returned

$$\lambda_1 \approx 1.00000000001$$

and

$$\lambda_2 \approx 2.000000000039,$$

which agrees with the expected analytical spectrum.

For the nonuniform density

$$\rho(x) = 1 + e^{10x},$$

I calculated and plotted the first 20 eigenvalues and their corresponding eigenfunctions. The spectrum was approximately linear over the first 20 modes, while the eigenfunctions became increasingly oscillatory toward the high-density end of the string.

Using an initial guess of

$$\lambda_0 = 0.5,$$

the shooting method converged to

$$\lambda \approx 0.573620,$$

which corresponds to the 17th eigenvalue of the calculated spectrum.

See `problem2.ipynb` or `problem2.pdf`.